

Inside India's Quick Commerce Revolution: A Data-Driven EDA of Customer Behavior & Operational Performance

Introduction

India's quick commerce industry is growing at an incredible pace. Customers today expect groceries and essentials to be delivered within minutes, while companies compete aggressively on delivery speed, customer experience, discounts, and operational efficiency.

Platforms such as Blinkit, Zepto, Swiggy Instamart, Dunzo, and BigBasket are continuously trying to improve delivery operations, increase customer retention, and strengthen their market presence across major cities.

But in an industry where speed and convenience are everything, several important business questions naturally arise:

- Does faster delivery actually improve customer satisfaction?
- Which cities contribute the most toward quick commerce growth?
- Which product categories dominate customer demand?
- Do discounts genuinely improve customer ordering behavior?
- What operational factors influence platform performance the most?
- Are customers prioritizing speed, pricing, or overall experience?

This Exploratory Data Analysis (EDA) is focused on understanding the operational and customer-side dynamics of India's quick commerce ecosystem using a large-scale dataset containing transactional, delivery, customer, and platform-related information.

Objective of This EDA

The goal of this analysis is not just to create visualizations, but to understand the business story hidden inside the data.

Through this EDA, the focus is on identifying:

- Customer behavior patterns
- Operational efficiency trends
- Delivery performance insights
- Platform competitiveness

- City-level demand patterns
- Revenue-driving factors
- Customer satisfaction indicators

The analysis aims to approach the dataset from a business intelligence perspective by uncovering insights that can support operational improvements, customer experience optimization, and strategic decision-making within the quick commerce industry.

About the Dataset

The dataset contains large-scale operational and transactional information related to multiple quick commerce platforms operating across India.

It includes data related to:

- Orders and delivery operations
- Product categories
- Delivery time and distance
- Discounts and payment methods
- Customer ratings and feedback
- City-level operations
- Platform performance metrics
- Delivery partner behavior

With nearly one million records, the dataset provides strong potential for exploring operational efficiency, customer trends, platform performance, and business growth patterns within the rapidly evolving quick commerce market.

Business Problem Statement

Quick commerce companies operate in a highly competitive environment where customer satisfaction, operational speed, and scalability directly influence business growth and long-term sustainability.

This analysis attempts to understand the key factors driving customer experience and operational performance while identifying patterns and insights that can help businesses optimize delivery operations, improve customer engagement, and strengthen overall platform performance.



Importing Required Libraries

Before starting the analysis, we first import the necessary Python libraries that will be used throughout this Exploratory Data Analysis (EDA).

These libraries will help in:

- Data manipulation and preprocessing

- Numerical computations
- Data visualization
- Statistical analysis
- Understanding patterns and trends within the dataset

The goal is to use these libraries to transform raw operational data into meaningful business insights related to India's quick commerce ecosystem.

Data Manipulation Libraries

```
import pandas as pd
```

```
import numpy as np
```

Data Visualization Libraries

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

Statistical & Warning Handling

```
import warnings
```

```
warnings.filterwarnings('ignore')
```

Loading the Dataset

The dataset is loaded into a Pandas DataFrame to begin the exploratory analysis process.

At this stage, the focus is on:

- Successfully importing the dataset
- Understanding the structure of the data
- Preparing for initial exploration and preprocessing

Loading the Dataset

```
df = pd.read_csv('quick_commerce_data_modified_cleaned.csv')
```

Displaying first 5 rows

```
df.head()
```

	Order_ID	Company	City	Customer_Age	Order_Value	Delivery_T
0	1000001	Swiggy Instamart	Noida	46	702	19
1	1000002	Flipkart Minutes	Amritsar	56	1007	20
2	1000003	Flipkart Minutes	Mumbai	18	1212	17
3	1000004	Swiggy Instamart	Delhi	23	1179	6
4	1000005	Dunzo	Mumbai	44	586	12

Understanding the Dataset Structure

Before jumping into analysis and visualizations, it is important to understand the structure and quality of the dataset.

In this section, we will explore:

- Shape of the dataset
- Column names
- Data types
- Missing values
- Statistical summary
- Initial observations

This helps in building a strong foundation for further analysis and ensures a better understanding of the operational and business variables present in the dataset.

Shape of the Dataset

```
print("Dataset Shape:", df.shape)
```

Dataset Shape: (947752, 13)

Displaying Column Names

```
df.columns
```

```
Index(['Order_ID', 'Company', 'City', 'Customer_Age', 'Order_Value',  
      'Delivery_Time_Min', 'Distance_Km', 'Items_Count', 'Product_Category',  
      'Payment_Method', 'Customer_Rating', 'Discount_Applied',  
      'Delivery_Partner_Rating'],  
      dtype='object')
```

Checking Dataset Information

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 947752 entries, 0 to 947751  
Data columns (total 13 columns):  
#   Column                               Non-Null Count  Dtype  
---  -  
0   Order_ID                             947752 non-null  int64  
1   Company                             947752 non-null  object  
2   City                                 947752 non-null  object  
3   Customer_Age                         947752 non-null  int64  
4   Order_Value                         947752 non-null  int64  
5   Delivery_Time_Min                   947752 non-null  int64  
6   Distance_Km                         947752 non-null  float64  
7   Items_Count                         947752 non-null  int64  
8   Product_Category                    947752 non-null  object  
9   Payment_Method                      947752 non-null  object  
10  Customer_Rating                     947752 non-null  int64  
11  Discount_Applied                    947752 non-null  int64  
12  Delivery_Partner_Rating              947752 non-null  int64
```

```
dtypes: float64(1), int64(8), object(4)
memory usage: 94.0+ MB
```

```
# Checking Missing Values
```

```
df.isnull().sum()
```

```
Order_ID          0
Company           0
City              0
Customer_Age      0
Order_Value       0
Delivery_Time_Min 0
Distance_Km       0
Items_Count       0
Product_Category  0
Payment_Method    0
Customer_Rating   0
Discount_Applied  0
Delivery_Partner_Rating 0
dtype: int64
```

```
# Statistical Summary of Numerical Columns
```

```
df.describe()
```

	Order_ID	Customer_Age	Order_Value	Delivery_Time_Min	
count	9.477520e+05	947752.000000	947752.000000	947752.000000	947752
mean	1.500028e+06	38.471165	570.912008	16.506620	72.118982
std	2.887273e+05	12.118982	374.175332	6.340851	4.000000
min	1.000001e+06	18.000000	50.000000	5.000000	0.000000
25%	1.249939e+06	28.000000	288.000000	12.000000	4.000000
50%	1.500026e+06	38.000000	532.000000	16.000000	7.000000
75%	1.750145e+06	49.000000	797.000000	20.000000	11.000000
max	2.000000e+06	59.000000	2492.000000	40.000000	11.000000

```
# Displaying Last 5 Rows
```

```
df.tail()
```

	Order_ID	Company	City	Customer_Age	Order_Value	Delivery_Time_Min
947747	1999996	Big Basket	Mumbai	48	72	18
947748	1999997	Swiggy Instamart	Mumbai	33	1510	18
947749	1999998	Jio Mart	Noida	29	637	18
947750	1999999	Dunzo	Pune	42	1103	13
947751	2000000	Amazon Now	Chennai	19	759	18



Initial Observations

After performing the initial exploration of the dataset, several important observations can already be identified from both the structural and statistical overview of the data.

Dataset Scale & Structure

- The dataset contains **947,752 records** with **13 columns**, indicating a large-scale operational dataset suitable for detailed business and customer behavior analysis.
 - The presence of nearly one million records provides strong analytical depth for identifying operational patterns, customer trends, and platform-level performance insights.
-



Data Quality Assessment

- The dataset appears to be **fully cleaned and structured**, with:
 - No missing values
 - Consistent data types
 - Properly formatted numerical and categorical columns
 - This is particularly important because it allows the analysis to focus more on extracting insights rather than spending excessive time on data cleaning and preprocessing.
-

Business & Operational Variables Present

The dataset includes a strong mix of:



Numerical Features

- Order Value
- Delivery Time
- Distance Covered
- Number of Items Ordered
- Customer Ratings
- Delivery Partner Ratings

Categorical Features

- Company
- City
- Product Category
- Payment Method

This combination creates strong potential for both:

- operational analysis
 - customer behavior analysis
-

Delivery & Operational Insights (Initial View)

From the statistical summary:

- The average delivery time is approximately **16.5 minutes**, which aligns closely with the quick commerce business model focused on rapid deliveries.
 - The average delivery distance is around **7.75 km**, suggesting that most deliveries operate within relatively short urban delivery zones.
 - Order values show noticeable variation, indicating the possibility of identifying:
 - high-value customer segments
 - category-level spending behavior
 - city-level purchasing trends
-

Customer Experience Indicators

- The dataset includes both:
 - Customer Ratings
 - Delivery Partner Ratings

This opens up opportunities to explore:

- customer satisfaction patterns
 - delivery efficiency impact
 - service quality relationships
 - operational bottlenecks affecting user experience
-

Business Intelligence Potential

Based on the available features, the dataset appears highly suitable for exploring business-critical questions such as:

- Which factors influence customer satisfaction?
- How does delivery performance impact ratings?
- Which cities and categories drive the highest operational activity?
- What operational variables influence order value?
- Which platforms perform better across different business metrics?

The dataset structure indicates strong potential for deriving meaningful operational and strategic insights related to India's growing quick commerce ecosystem.

Beginning the Exploratory Analysis

The objective is to understand:

- customer behavior
- operational efficiency
- delivery performance
- platform competitiveness
- and factors influencing customer satisfaction

through meaningful analysis and visualization.

Does Faster Delivery Improve Customer Ratings?

Delivery speed is one of the biggest competitive advantages in the quick commerce industry. Companies continuously invest in logistics, dark stores, and operational optimization to reduce delivery times and improve customer experience.

But an important business question still remains:

Are faster deliveries actually resulting in better customer ratings?

Understanding this relationship is important because customer satisfaction directly impacts:

- customer retention
- repeat orders
- brand trust
- and overall platform competitiveness.

In this analysis, we will explore the relationship between delivery time and customer ratings to understand whether delivery efficiency is influencing customer experience.

Creating Delivery Time Bins

```
df['Delivery_Time_Bins'] = pd.cut(
    df['Delivery_Time_Min'],
    bins=[0, 10, 20, 30, 40, 50, 60],
    labels=['0-10', '10-20', '20-30', '30-40', '40-50', '50-60']
)
```

Average Ratings by Delivery Time Range

```
avg_rating = df.groupby('Delivery_Time_Bins')
               ['Customer_Rating'].mean().reset_index()
```

Plotting

```
plt.figure(figsize=(10,6))
```

```
sns.barplot(
```



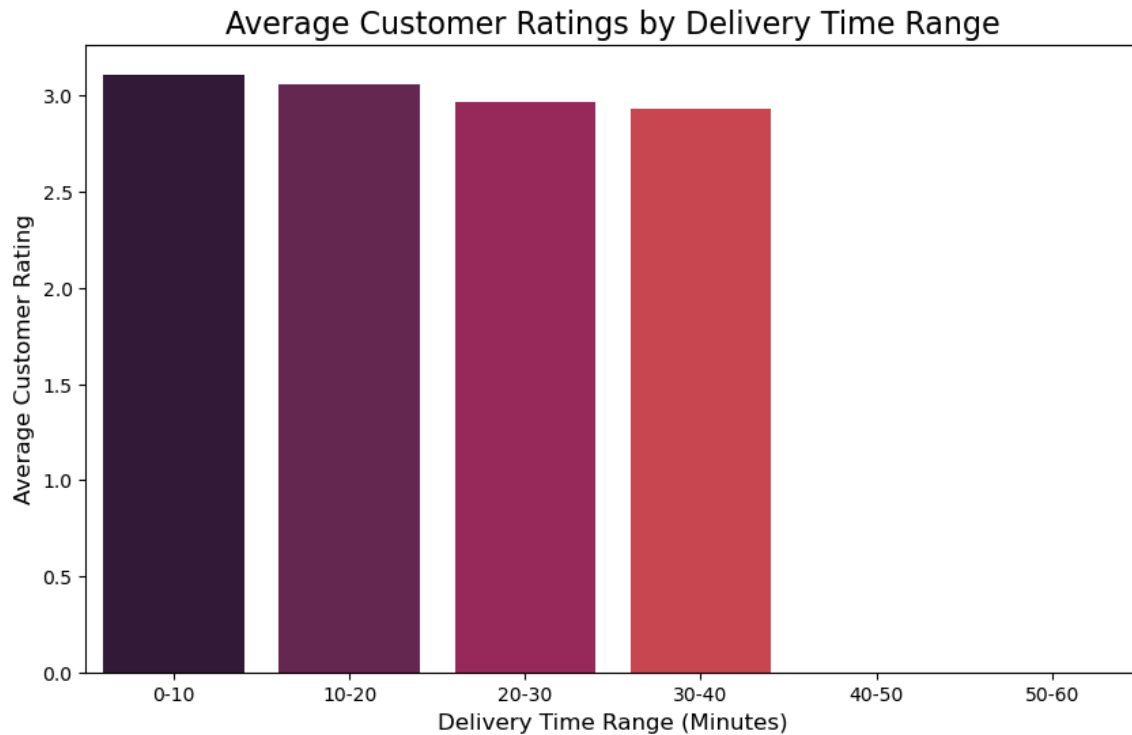
```

x='Delivery_Time_Bins',
y='Customer_Rating',
data=avg_rating,
palette='rocket'
)

plt.title('Average Customer Ratings by Delivery Time Range', fontsize=16)
plt.xlabel('Delivery Time Range (Minutes)', fontsize=12)
plt.ylabel('Average Customer Rating', fontsize=12)

plt.show()

```



Observation

Customer ratings show a gradual decline as delivery time increases. Orders delivered within 0–10 minutes received the highest average ratings, while longer delivery durations saw slightly lower customer satisfaction.

The trend suggests that faster deliveries positively influence customer experience, reinforcing the importance of delivery efficiency in quick commerce operations.

Which Cities Experience the Highest Average Delivery Time?

Identifying cities with higher average delivery times can help businesses:

- detect operational bottlenecks

- optimize delivery networks
- improve logistics planning
- and prioritize city-level operational improvements.

This analysis focuses on understanding how delivery performance varies across different cities within the quick commerce ecosystem.

Average Delivery Time by City

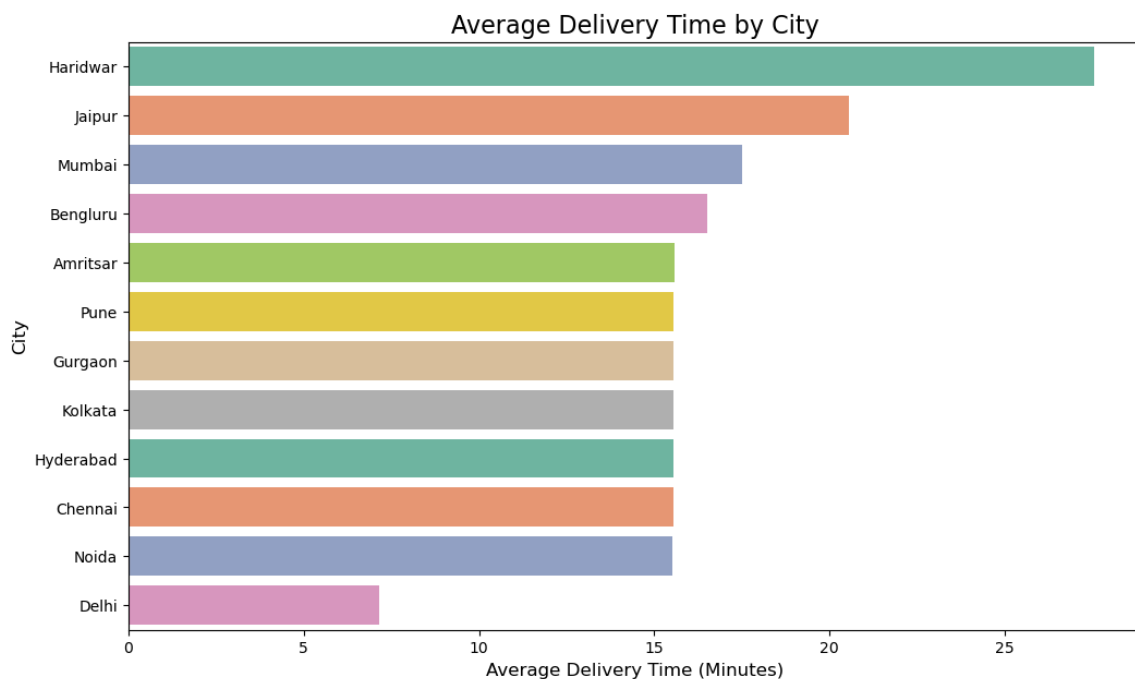
```
city_delivery_time = df.groupby('City')
                        ['Delivery_Time_Min'].mean().sort_values(ascending=False)
```

```
plt.figure(figsize=(12,7))
```

```
sns.barplot(
    x=city_delivery_time.values,
    y=city_delivery_time.index,
    palette='Set2'
)
```

```
plt.title('Average Delivery Time by City', fontsize=16)
plt.xlabel('Average Delivery Time (Minutes)', fontsize=12)
plt.ylabel('City', fontsize=12)
```

```
plt.show()
```



Observation

Delivery performance varies significantly across cities. Haridwar recorded the highest average delivery time, followed by Jaipur and Mumbai, while Delhi showed the lowest average delivery duration among all cities.

The variation suggests that operational efficiency is not uniform across locations and may be influenced by factors such as traffic conditions, delivery demand, infrastructure, or order density.

Cities with consistently higher delivery times may require:

- better delivery optimization
- stronger logistics support
- or improved operational planning

to maintain customer satisfaction and delivery efficiency.

Do Discounts Increase Customer Spending?

Discounts are widely used in quick commerce platforms to attract customers and encourage purchases. However, businesses also need to understand whether discounts are genuinely increasing customer spending or simply reducing profitability.

This analysis focuses on comparing average order values between:

- orders where discounts were applied
- and orders without discounts

to understand whether discounts influence customer purchasing behavior and spending patterns.

```
# Average Order Value based on Discount Availability
```

```
discount_order_value = df.groupby('Discount_Applied')  
    ['Order_Value'].mean().reset_index()
```

```
# Renaming Labels
```

```
discount_order_value['Discount_Applied'] =  
    discount_order_value['Discount_Applied'].replace({  
        0: 'No Discount',  
        1: 'Discount Applied'  
    })
```

```
# Plotting
```

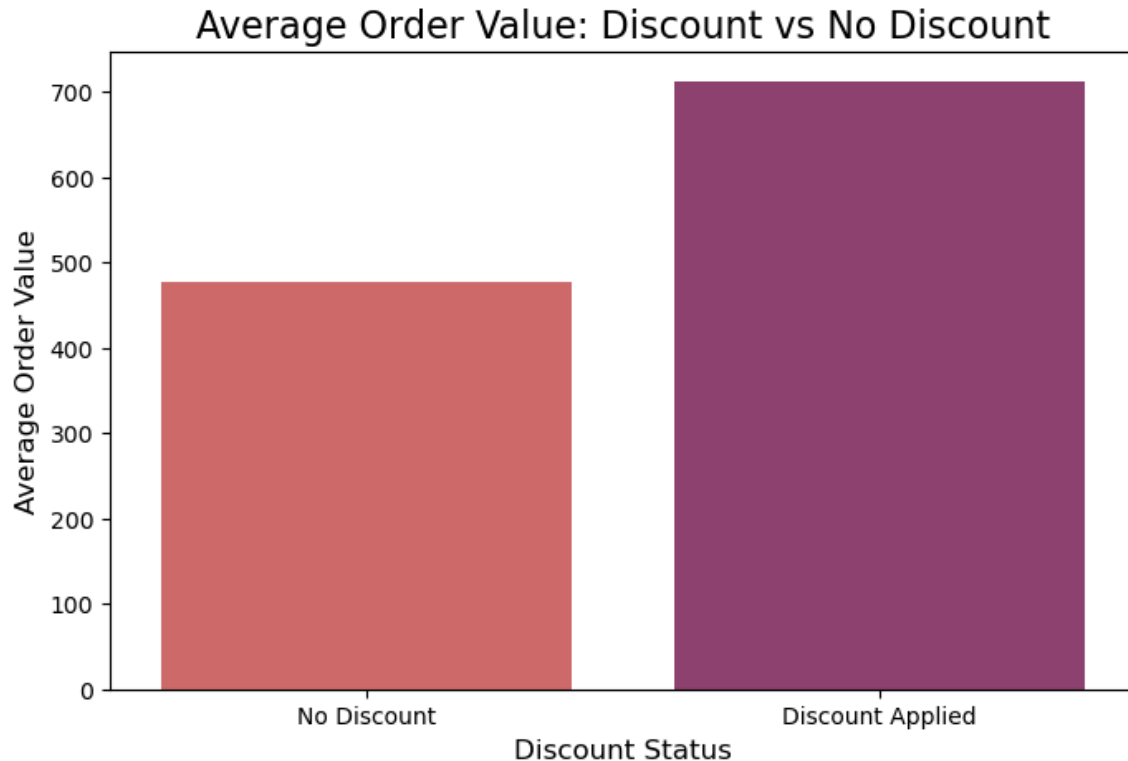
```
plt.figure(figsize=(8,5))
```

```
sns.barplot(  
    x='Discount_Applied',  
    y='Order_Value',  
    data=discount_order_value,  
    palette='flare'  
)
```

```
plt.title('Average Order Value: Discount vs No Discount', fontsize=16)
```

```
plt.xlabel('Discount Status', fontsize=12)
plt.ylabel('Average Order Value', fontsize=12)

plt.show()
```



Observation

Orders where discounts were applied showed noticeably higher average order values compared to orders without discounts.

This suggests that discounts are positively influencing customer spending behavior, encouraging customers to place higher-value orders during promotional purchases.

The insight indicates that discount strategies may not only attract customers but also increase basket size and overall transaction value within quick commerce platforms.

Which Cities Generate the Highest Average Order Value?

Customer purchasing behavior can vary significantly across cities due to factors such as:

- purchasing power
- customer demand
- platform penetration
- lifestyle patterns
- and spending habits.

Identifying cities with higher average order values can help businesses:

- prioritize expansion strategies
- optimize city-level marketing
- improve inventory allocation
- and identify high-value operational markets.

```
# Average Order Value by City
```

```
city_order_value = df.groupby('City')  
                    ['Order_Value'].mean().sort_values(ascending=False)
```

```
plt.figure(figsize=(12,7))
```

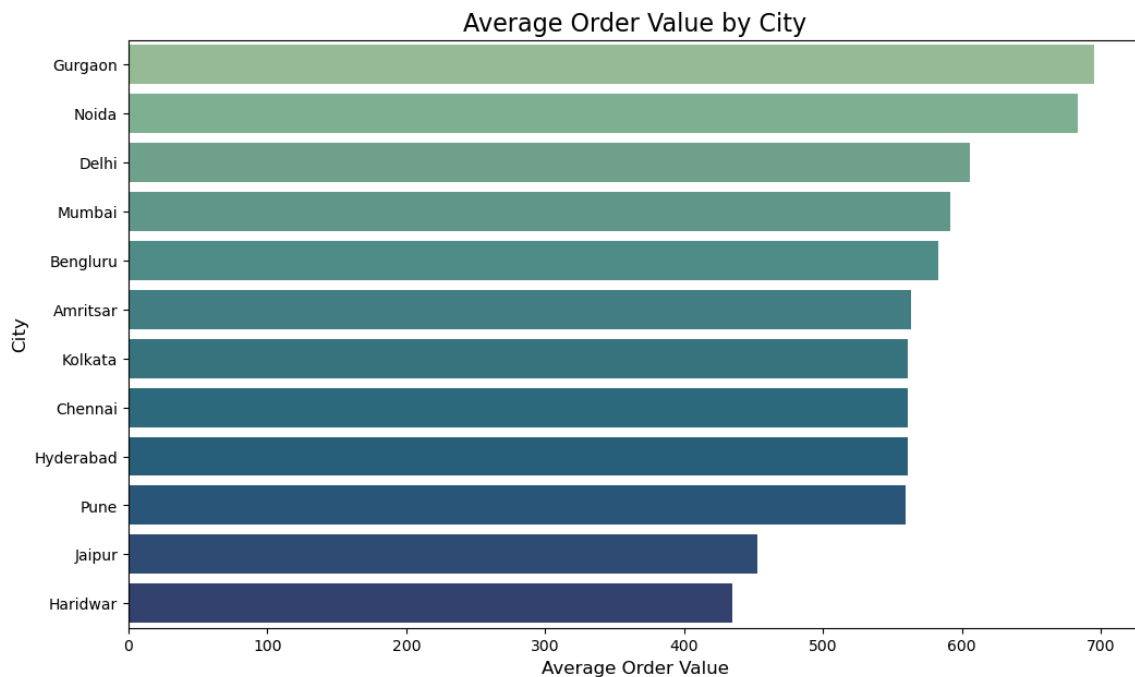
```
sns.barplot(  
    x=city_order_value.values,  
    y=city_order_value.index,  
    palette='crest'  
)
```

```
plt.title('Average Order Value by City', fontsize=16)
```

```
plt.xlabel('Average Order Value', fontsize=12)
```

```
plt.ylabel('City', fontsize=12)
```

```
plt.show()
```



Observation

Average order value varies noticeably across cities. Gurgaon and Noida recorded the highest average order values, while Jaipur and Haridwar showed comparatively lower customer spending patterns.

The trend suggests that customer purchasing behavior and spending capacity differ significantly across locations, potentially influenced by factors such as urban demand, purchasing power, and platform penetration.

Higher-value cities may represent stronger revenue opportunities for quick commerce platforms and could be prioritized for operational expansion, premium offerings, or targeted marketing strategies.

Which Quick Commerce Platform Delivers the Fastest?

Delivery speed is one of the biggest competitive advantages in the quick commerce industry. Platforms continuously compete to reduce delivery time and improve operational efficiency in order to attract and retain customers.

Comparing delivery performance across platforms can help identify:

- operational leaders
- logistics efficiency
- fulfillment capability
- and overall service competitiveness within the market.

Average Delivery Time by Platform

```
platform_delivery = df.groupby('Company')
                    ['Delivery_Time_Min'].mean().sort_values()

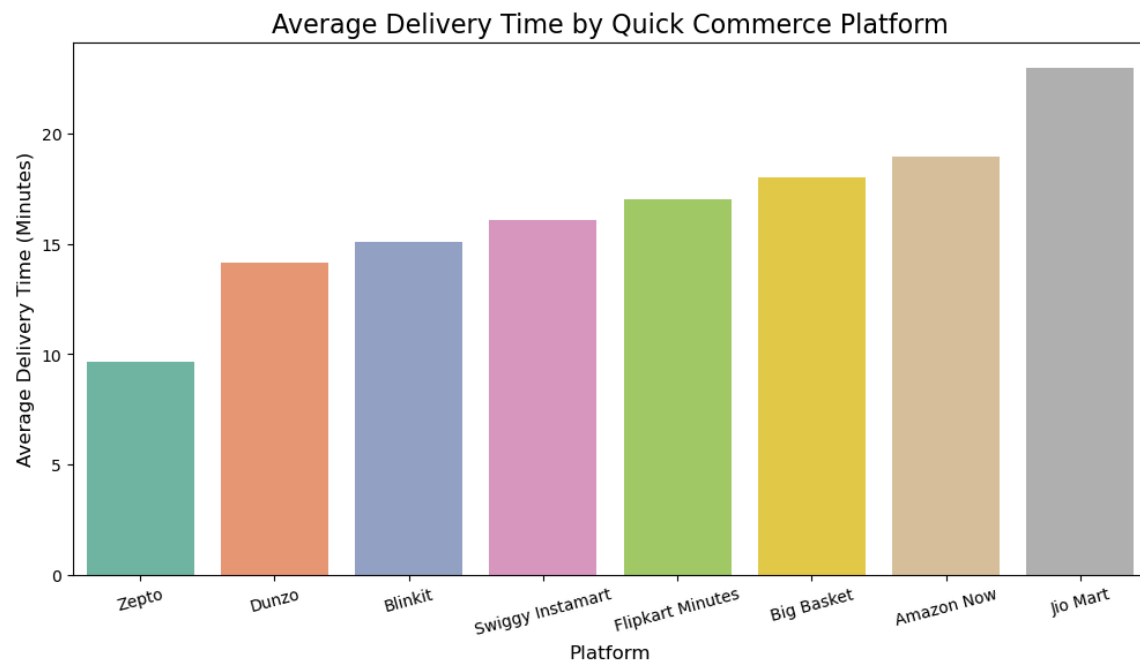
plt.figure(figsize=(12,6))

sns.barplot(
    x=platform_delivery.index,
    y=platform_delivery.values,
    palette='Set2'
)

plt.title('Average Delivery Time by Quick Commerce Platform', fontsize=16)
plt.xlabel('Platform', fontsize=12)
plt.ylabel('Average Delivery Time (Minutes)', fontsize=12)

plt.xticks(rotation=15)

plt.show()
```



Observation

Delivery performance varies significantly across quick commerce platforms. Zepto recorded the fastest average delivery time among all platforms, while Jio Mart showed the slowest delivery performance.

Platforms such as Blinkit, Swiggy Instamart, and Dunzo maintained relatively competitive delivery timelines, whereas Amazon Now and Big Basket appeared comparatively slower in fulfilling orders.

The variation highlights noticeable differences in operational efficiency, logistics execution, and delivery infrastructure across platforms operating within the quick commerce ecosystem.

Which Quick Commerce Platform Has the Highest Customer Satisfaction?

In the quick commerce industry, customer satisfaction plays a critical role in customer retention, repeat purchases, and long-term platform growth.

While delivery speed is important, customer ratings help measure how users actually perceive their overall experience with a platform.

Average Customer Ratings by Platform

```
platform_rating = df.groupby('Company')  
                    ['Customer_Rating'].mean().sort_values(ascending=False)
```

```
plt.figure(figsize=(12,6))
```

```

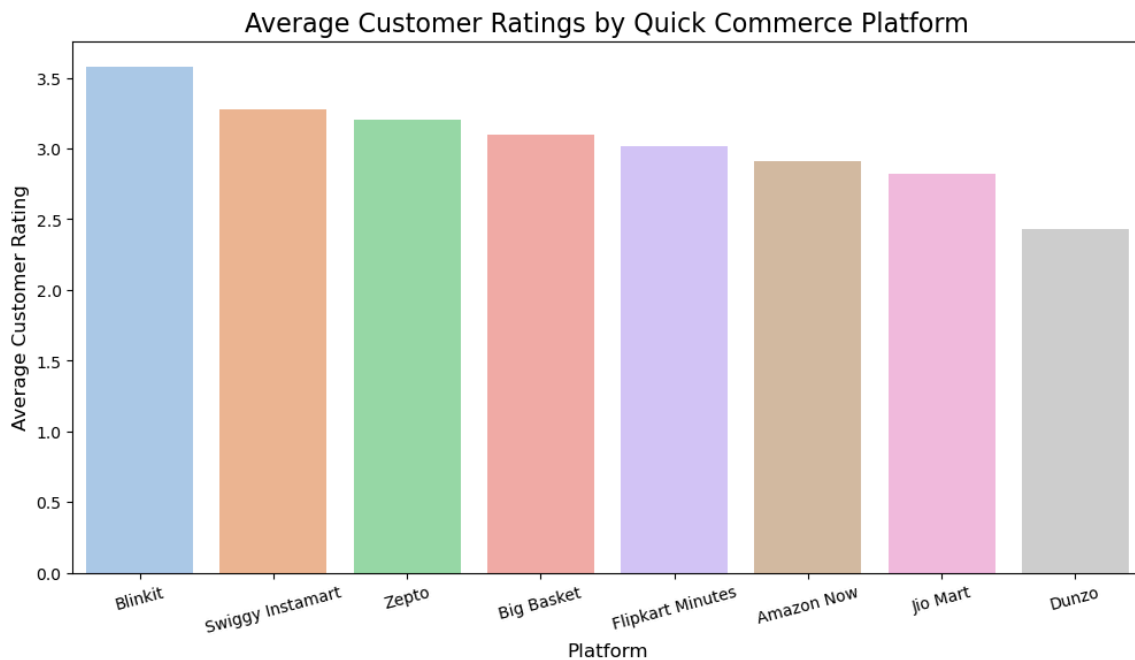
sns.barplot(
    x=platform_rating.index,
    y=platform_rating.values,
    palette='pastel'
)

plt.title('Average Customer Ratings by Quick Commerce Platform', fontsize=16)
plt.xlabel('Platform', fontsize=12)
plt.ylabel('Average Customer Rating', fontsize=12)

plt.xticks(rotation=15)

plt.show()

```



Observation

Customer satisfaction varies noticeably across quick commerce platforms. Blinkit recorded the highest average customer rating, while Dunzo received the lowest overall customer satisfaction scores.

Interestingly, the results also suggest that faster delivery alone does not completely determine customer satisfaction. Although Zepto showed the fastest delivery performance in earlier analysis, Blinkit achieved higher customer ratings overall.

This indicates that customer experience is likely influenced by multiple factors beyond delivery speed, including service quality, order accuracy, platform experience, and overall operational reliability.

How Do Quick Commerce Platforms Perform Across Different Cities?

Quick commerce platforms do not always perform equally across every city. Factors such as customer demand, operational infrastructure, regional penetration, and logistics efficiency can influence platform performance across locations.

Understanding how platforms perform in different cities can help identify:

- dominant market players
- high-performing operational regions
- competitive positioning
- and potential expansion opportunities.

Average Order Value by City and Platform

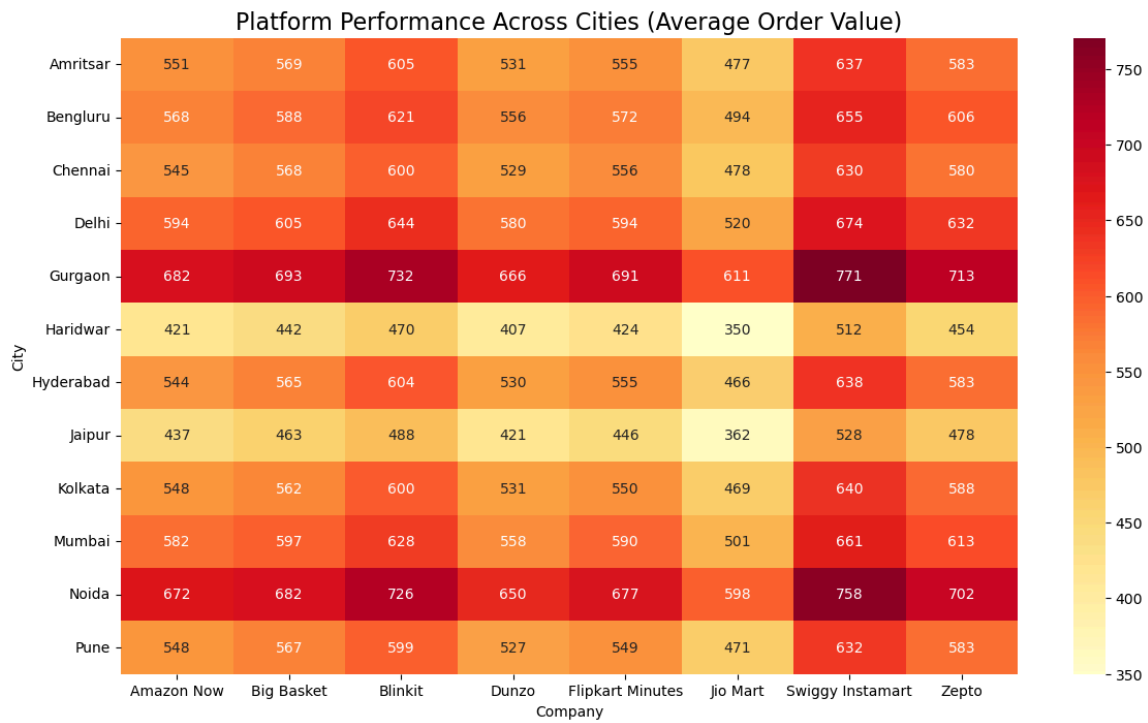
```
city_platform = df.pivot_table(
    values='Order_Value',
    index='City',
    columns='Company',
    aggfunc='mean'
)

plt.figure(figsize=(14,8))

sns.heatmap(
    city_platform,
    annot=True,
    fmt='.0f',
    cmap='YlOrRd'
)

plt.title('Platform Performance Across Cities (Average Order Value)',
        fontsize=16)

plt.show()
```



Average Order Value by Platform

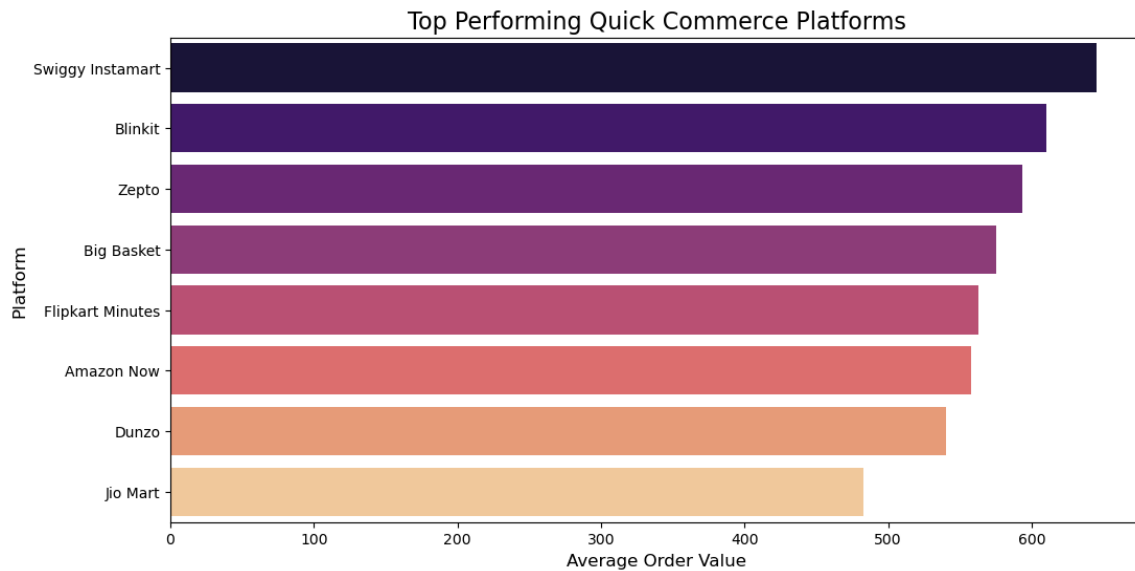
```
platform_order = df.groupby('Company')
                    ['Order_Value'].mean().sort_values(ascending=False)

plt.figure(figsize=(12,6))

sns.barplot(
    x=platform_order.values,
    y=platform_order.index,
    palette='magma'
)

plt.title('Top Performing Quick Commerce Platforms', fontsize=16)
plt.xlabel('Average Order Value', fontsize=12)
plt.ylabel('Platform', fontsize=12)

plt.show()
```



Observation

Swiggy Instamart emerged as the top-performing quick commerce platform based on average order value, followed closely by Blinkit and Zepto. On the other hand, Jio Mart and Dunzo recorded comparatively lower average order values across the dataset.

The variation suggests noticeable differences in customer spending patterns, platform penetration, and overall purchasing behavior across platforms.

The results indicate that leading platforms may be attracting higher-value transactions through stronger operational presence, product availability, customer trust, or better overall platform experience.

How Does Delivery Distance Impact Delivery Time?

In quick commerce operations, delivery efficiency is heavily influenced by distance. As the delivery radius increases, platforms may experience longer fulfillment times due to traffic conditions, routing complexity, and operational constraints.

Understanding the relationship between delivery distance and delivery time can help businesses:

- optimize delivery coverage
- improve logistics planning
- reduce operational delays
- and maintain faster fulfillment performance.

Creating Distance Groups

```
df['Distance_Group'] = pd.cut(  
    df['Distance_Km'],  
    bins=[0, 3, 6, 9, 12, 15],
```

```

    labels=['0-3 km', '3-6 km', '6-9 km', '9-12 km', '12-15 km']
)

# Average Delivery Time by Distance Group

distance_delivery = df.groupby('Distance_Group')
    ['Delivery_Time_Min'].mean().reset_index()

# Plotting

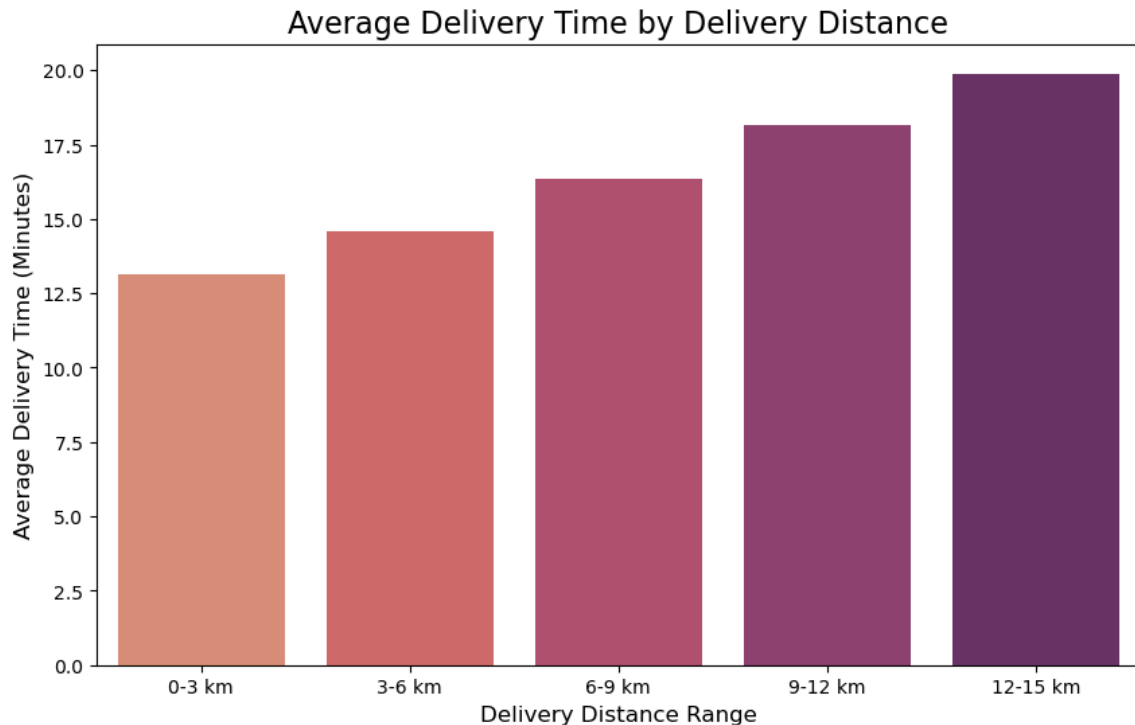
plt.figure(figsize=(10,6))

sns.barplot(
    x='Distance_Group',
    y='Delivery_Time_Min',
    data=distance_delivery,
    palette='flare'
)

plt.title('Average Delivery Time by Delivery Distance', fontsize=16)
plt.xlabel('Delivery Distance Range', fontsize=12)
plt.ylabel('Average Delivery Time (Minutes)', fontsize=12)

plt.show()

```



Observation

The analysis shows a clear upward trend between delivery distance and delivery time. Orders delivered within shorter distances (0–3 km) recorded the fastest fulfillment times, while deliveries covering longer distances consistently required more time.

The steady increase in delivery duration across distance ranges highlights the direct operational impact of delivery radius on quick commerce efficiency.

This suggests that optimizing delivery coverage, improving dark store placement, and strengthening route efficiency could play a critical role in maintaining faster delivery performance as delivery distances increase.

Which Product Categories Generate Higher Order Values?

Different product categories contribute differently to customer spending behavior within quick commerce platforms.

Understanding which categories generate higher order values can help businesses:

- identify high-revenue segments
- optimize inventory allocation
- improve category-focused marketing
- and prioritize operational resources toward high-value product segments.

This analysis focuses on comparing average order value across different product categories.

```
# Total Orders by Product Category
```

```
category_orders = df['Product_Category'].value_counts()
```

```
# Plotting
```

```
plt.figure(figsize=(12,6))
```

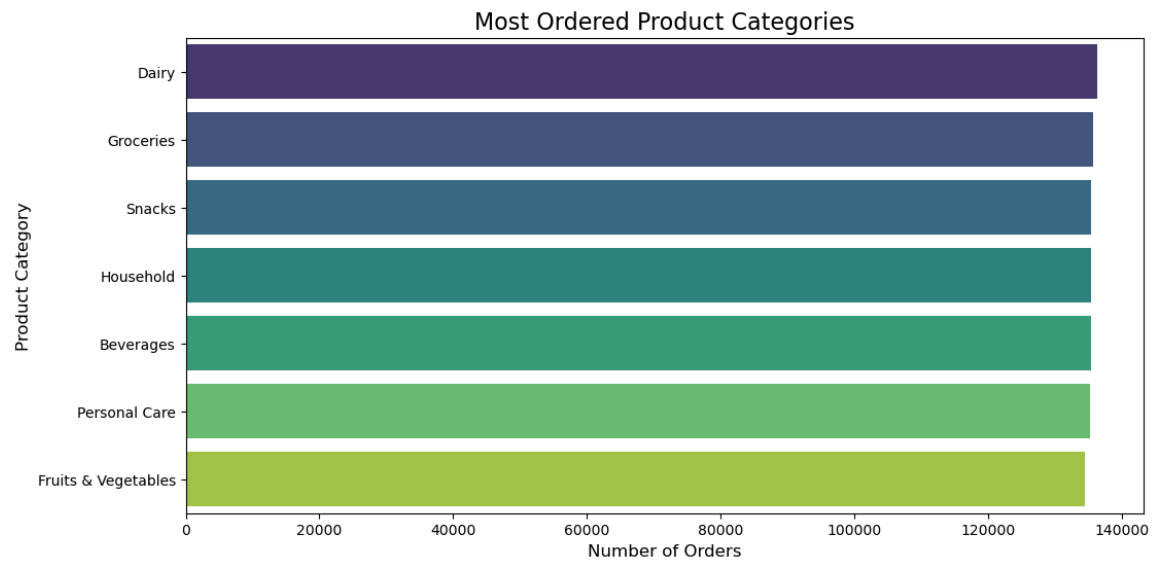
```
sns.barplot(  
    x=category_orders.values,  
    y=category_orders.index,  
    palette='viridis'  
)
```

```
plt.title('Most Ordered Product Categories', fontsize=16)
```

```
plt.xlabel('Number of Orders', fontsize=12)
```

```
plt.ylabel('Product Category', fontsize=12)
```

```
plt.show()
```



City-Level Operational Metrics

```
city_risk = df.groupby('City').agg({
    'Delivery_Time_Min': 'mean',
    'Customer_Rating': 'mean',
    'Order_Value': 'mean'
}).reset_index()
```

city_risk

	City	Delivery_Time_Min	Customer_Rating	Order_Value
0	Amritsar	15.572865	3.031054	563.590688
1	Bengluru	16.518981	3.473977	582.751422
2	Chennai	15.537591	3.033137	560.601222
3	Delhi	7.140699	3.123077	605.299298
4	Gurgaon	15.558390	3.045451	694.917390
5	Haridwar	27.547815	3.036522	434.928565
6	Hyderabad	15.539503	2.506569	560.574879
7	Jaipur	20.553798	2.941303	452.645533
8	Kolkata	15.541177	3.029018	561.074205
9	Mumbai	17.517304	3.217668	591.369940
10	Noida	15.518283	3.040346	683.324658
11	Pune	15.563463	3.036661	559.732233

#Now Let's create:

a weighted efficiency score.

Platform-Level Metrics

```
platform_metrics = df.groupby('Company').agg({
    'Delivery_Time_Min': 'mean',
```

```

        'Customer_Rating': 'mean',
        'Order_Value': 'mean'
    }).reset_index()

```

platform_metrics

	Company	Delivery_Time_Min	Customer_Rating	Order_Value
0	Amazon Now	18.984155	2.913417	557.510429
1	Big Basket	18.027345	3.097902	575.065542
2	Blinkit	15.121771	3.580762	609.819198
3	Dunzo	14.177060	2.428024	540.317053
4	Flipkart Minutes	17.045080	3.016316	563.176472
5	Jio Mart	23.013224	2.819747	482.914592
6	Swiggy Instamart	16.072690	3.279443	644.927250
7	Zepto	9.644313	3.203062	593.258314

```

from sklearn.preprocessing import MinMaxScaler

```

```

scaler = MinMaxScaler()

```

```

# Normalizing Metrics

```

```

platform_metrics[['Delivery_Norm', 'Rating_Norm', 'Order_Norm']] =
    scaler.fit_transform(
        platform_metrics[['Delivery_Time_Min', 'Customer_Rating', 'Order_Value']]
    )

```

platform_metrics

	Company	Delivery_Time_Min	Customer_Rating	Order_Value	Delivery
0	Amazon Now	18.984155	2.913417	557.510429	0.698624
1	Big Basket	18.027345	3.097902	575.065542	0.627054
2	Blinkit	15.121771	3.580762	609.819198	0.409716
3	Dunzo	14.177060	2.428024	540.317053	0.339051
4	Flipkart Minutes	17.045080	3.016316	563.176472	0.553580
5	Jio Mart	23.013224	2.819747	482.914592	1.000000
6	Swiggy Instamart	16.072690	3.279443	644.927250	0.480845
7	Zepto	9.644313	3.203062	593.258314	0.000000

```

# Creating Platform Efficiency Score

```

```
platform_metrics['Efficiency_Score'] = (
    (1 - platform_metrics['Delivery_Norm']) * 0.4 +
    platform_metrics['Rating_Norm'] * 0.3 +
    platform_metrics['Order_Norm'] * 0.3
)
```

Sorting Platforms

```
platform_efficiency = platform_metrics.sort_values(
    by='Efficiency_Score',
    ascending=False
)
```

platform_efficiency

	Company	Delivery_Time_Min	Customer_Rating	Order_Value	Delivery
7	Zepto	9.644313	3.203062	593.258314	0.000000
2	Blinkit	15.121771	3.580762	609.819198	0.409716
6	Swiggy Instamart	16.072690	3.279443	644.927250	0.480845
1	Big Basket	18.027345	3.097902	575.065542	0.627054
4	Flipkart Minutes	17.045080	3.016316	563.176472	0.553580
0	Amazon Now	18.984155	2.913417	557.510429	0.698624
3	Dunzo	14.177060	2.428024	540.317053	0.339051
5	Jio Mart	23.013224	2.819747	482.914592	1.000000

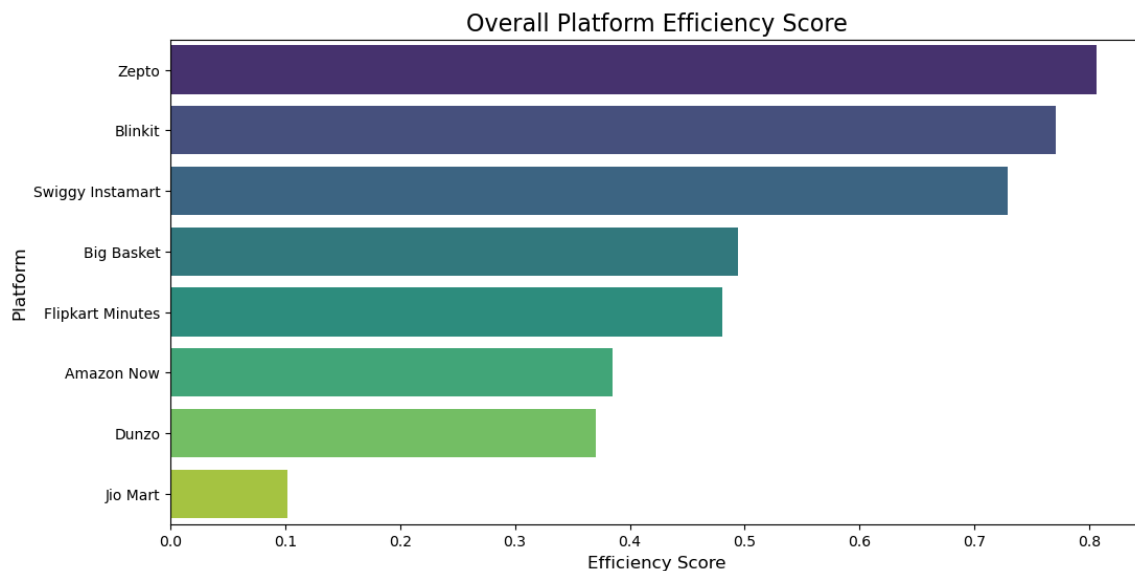
Plotting Platform Efficiency Score

```
plt.figure(figsize=(12,6))
```

```
sns.barplot(
    data=platform_efficiency,
    x='Efficiency_Score',
    y='Company',
    palette='viridis'
)
```

```
plt.title('Overall Platform Efficiency Score', fontsize=16)
plt.xlabel('Efficiency Score', fontsize=12)
plt.ylabel('Platform', fontsize=12)
```

```
plt.show()
```

Observation

The custom Platform Efficiency Score highlights Zepto as the strongest overall performer across the quick commerce ecosystem, driven by its faster delivery performance and competitive customer satisfaction metrics.

- Blinkit and Swiggy Instamart also demonstrated strong operational efficiency, indicating a balanced combination of delivery speed, customer experience, and order value generation.
- On the other hand, platforms such as Jio Mart and Dunzo recorded significantly lower efficiency scores, suggesting potential operational inefficiencies and weaker overall performance across key business dimensions.

The analysis demonstrates how combining multiple KPIs into a unified performance framework can provide a more strategic evaluation of platform competitiveness compared to analyzing isolated metrics individually.

Which Cities Generate High Revenue but Face Delivery Inefficiencies?

In quick commerce, high-performing cities are expected to maintain both:

- strong customer spending
- and efficient delivery operations.

However, some cities may generate higher order values while simultaneously experiencing slower delivery performance. These cities represent critical operational pressure points because inefficient fulfillment in high-value markets can negatively impact customer experience and long-term growth.

Identifying such cities can help businesses:

- prioritize operational improvements
- optimize logistics infrastructure
- improve delivery efficiency in key revenue markets
- and support smarter expansion decisions.

City-Level Revenue & Delivery Analysis

```
city_analysis = df.groupby('City').agg({  
    'Order_Value': 'mean',  
    'Delivery_Time_Min': 'mean'  
}).reset_index()
```

Sorting by Order Value

```
city_analysis = city_analysis.sort_values(by='Order_Value', ascending=False)
```

Plotting

```
fig, ax1 = plt.subplots(figsize=(12,6))
```

Order Value Bars

```
sns.barplot(  
    x='City',  
    y='Order_Value',  
    data=city_analysis,  
    ax=ax1,  
    color='skyblue'  
)
```

```
ax1.set_ylabel('Average Order Value')  
ax1.set_xlabel('City')
```

Secondary Axis for Delivery Time

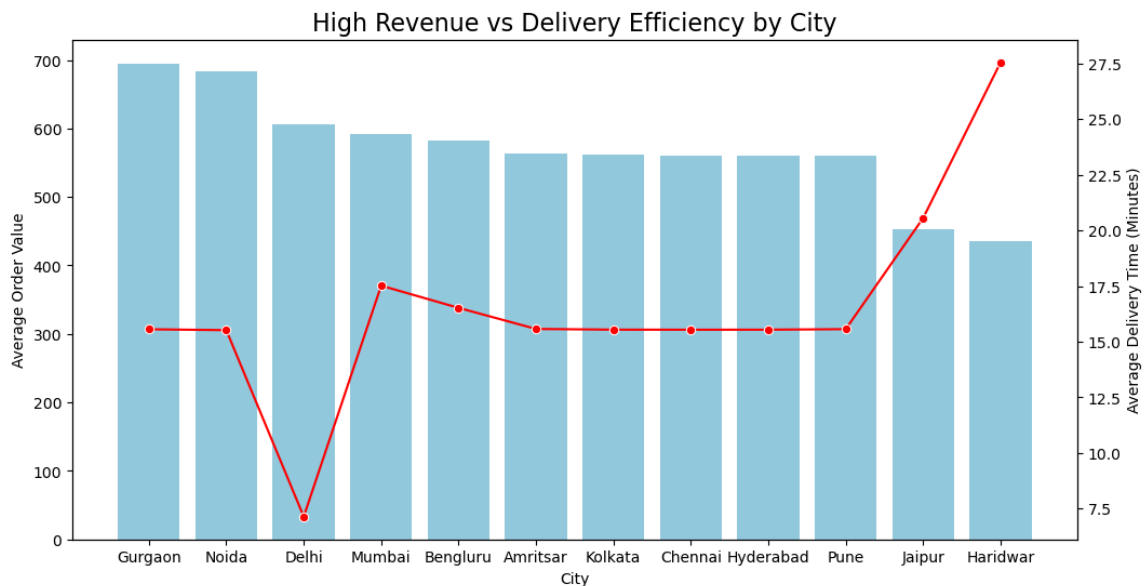
```
ax2 = ax1.twinx()
```

```
sns.lineplot(  
    x='City',  
    y='Delivery_Time_Min',  
    data=city_analysis,  
    marker='o',  
    sort=False,  
    ax=ax2,  
    color='red'  
)
```

```
ax2.set_ylabel('Average Delivery Time (Minutes)')
```

```
plt.title('High Revenue vs Delivery Efficiency by City', fontsize=16)
```

```
plt.show()
```



Observation

- Cities such as Gurgaon and Noida generated the highest average order values while still maintaining relatively stable delivery performance, indicating stronger operational balance and healthier market conditions.
- Delhi emerged as one of the most operationally efficient cities, combining strong customer spending with the lowest average delivery time. This suggests a well-optimized logistics network and stronger fulfillment efficiency within the region.
- On the other hand, cities such as Mumbai and Bengaluru showed strong revenue potential but comparatively higher delivery times, indicating possible operational pressure in high-demand markets. These cities may require logistics optimization, improved routing efficiency, or expanded fulfillment infrastructure to maintain customer experience as demand grows.
- Haridwar and Jaipur recorded both lower order values and significantly higher delivery times, highlighting weaker operational efficiency and lower commercial performance compared to larger metropolitan markets.

Overall, the analysis suggests that operational efficiency plays a critical role in sustaining customer experience and scalability within high-value quick commerce markets.

Final Conclusion

This analysis was focused on understanding how operational efficiency, customer experience, and platform performance interact within the quick commerce ecosystem.

While exploring the dataset, one clear pattern that consistently emerged was the impact of delivery efficiency on overall customer experience. Longer delivery distances led to increased delivery times, and cities facing slower fulfillment generally showed weaker operational performance compared to more optimized markets.

The analysis also highlighted that high-revenue cities such as Gurgaon, Noida, and Delhi represent strong growth markets for quick commerce platforms. However, certain cities with strong order potential also showed signs of operational inefficiency, suggesting that scaling demand without strengthening logistics infrastructure can eventually impact customer experience and fulfillment performance.

Platform-level comparisons showed that performance in quick commerce cannot be evaluated using a single KPI alone. Delivery speed, customer satisfaction, and order value often behave differently across platforms, making isolated analysis insufficient for strategic decision-making. To address this, a custom Platform Efficiency Score was developed by combining multiple operational and customer-focused metrics into a unified benchmarking framework.

One of the key takeaways from this project was understanding the importance of filtering out weak or non-actionable insights during exploratory analysis. Several visualizations were intentionally excluded because they did not contribute meaningful business value or realistic decision-making direction. This helped maintain analytical quality and kept the focus on insights that were operationally relevant and strategically useful.

Overall, the project demonstrates how exploratory analysis can move beyond descriptive visualization and evolve into business-focused analytical thinking by connecting logistics performance, customer behavior, and platform competitiveness into a more decision-oriented framework.